SUMMER TRAINING/INTERNSHIP REPORT

(Term Aug-Dec 2026)

STUDENT DROPOUT PREDICTION PROJECT

Submitted by:

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Course Code: PETV149

UNDER THE GUIDANCE OF

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School of Computer Science and Engineering

LOVELY PROFESSIONAL UNIVERSITY

Acknowledgement

We would like to express our sincere gratitude to our project guides and the faculty members for their invaluable guidance, constant supervision, and continuous support throughout the duration of our summer internship. Their encouragement and insightful feedback have been instrumental in the successful completion of our project titled **“Student Dropout Prediction Project – Education Analytics.”**

We are especially thankful to our friends and peers for providing us with the necessary feedback, information, and constructive suggestions that enabled us to enhance our knowledge and skills. Their suggestions have not only helped us achieve the objectives of this project but have also contributed significantly to our professional growth.

We also extend our heartfelt thanks to our institution and the Department of Computer Science and Engineering for giving us the opportunity to undertake this internship project and for creating an environment that fostered learning and innovation.

Finally, we wish to express our gratitude to our team members for the cooperation, dedication, and hard work they put into the successful completion of this project.

Team Outliers

- Harsh Soni – 12401684 (B. Tech CSE)
- Priyanshu Singh – 12527685 (MCA-[AI & ML])

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1. Introduction

The Department of Computer Science is dedicated to fostering academic excellence and practical innovation, providing students with the resources and guidance to solve complex, real-world problems. This project was carried out as part of the Summer Internship Course offered by **Ms. Savleen Kaur and Ms. Sandeep Kaur**, conducted **from 11th June 2026 to 19th July 2026.**

The internship simulates a real-world business scenario, treating the university as the client organisation. The objective was to apply the theoretical concepts of machine learning and data visualisation learned throughout the curriculum to a comprehensive dataset, mimicking the workflow and deliverables expected in a professional data analytics role.

1.1 Overview of Training Domain

This training programme focused on applying machine learning pipelines and engineering end-to-end predictive software tools to address real-world operational challenges in large-scale institutional databases. The work involved handling real-world data, applying statistical methods, building predictive models, and creating comprehensive dashboards for data visualisation and business intelligence.

1.2 Objective of the Project

Educational institutions experience significant operational inefficiencies and reduced completion metrics due to unpredicted student attrition. The objective of this project is to build an automated predictive engine that evaluates multi-dimensional parameters — including academic records (CGPA, GPA), behavioural markers (attendance rate, assignment delays), socio-economic indicators (family income, internet access), and demographics — to isolate early student dropout risks before formal withdrawal occurs.

2. Training Overview

2.1 Tools & Technologies Used

Programming & ML: Python, Pandas, NumPy, Scikit-learn (SimpleImputer, StandardScaler, OneHotEncoder, ColumnTransformer, Pipeline, Ensemble Classifiers)

Data Visualisation: Matplotlib, Seaborn, Microsoft Power BI, Power Query

Development Environment: Jupyter Notebook

2.2 Areas Covered During Training

- SQL
- Data Cleaning and Preprocessing
- Statistical Analysis and Hypothesis Testing
- Machine Learning & AI Automation
- Model Evaluation and Interpretation
- Business Intelligence and Dashboard Design with Power BI

2.3 Summary of Work

The project followed a structured, end-to-end analytics workflow. Work began with the ingestion of a 10,000-record institutional dataset covering demographic, academic, behavioural and socio-economic attributes, followed by systematic cleaning through scikit-learn Pipeline and ColumnTransformer objects that imputed missing values and standardised/encoded features without risk of data leakage. This was followed by an Exploratory Data Analysis (EDA) phase that used boxplots, histograms and count plots to uncover the strongest behavioural and academic predictors of dropout.

Three supervised classification algorithms — Logistic Regression, Random Forest and Gradient Boosting — were then trained on a stratified 80/20 train-test split and benchmarked using accuracy, ROC-AUC, precision and recall. Logistic Regression was selected as the production model on the strength of its accuracy and AUC scores. Finally, the trained model was used to score the entire student population, and the resulting dropout probabilities and risk tiers were exported to a CSV file engineered for direct consumption by an interactive Power BI dashboard.

3. Project Details

3.1 Title of the Project

Student Dropout Prediction Project – Education Analytics

3.2 Problem Definition

Design a predictive model that leverages student academic performance, attendance history, socio-economic background, and classroom engagement metrics to forecast dropout risk. Use classification algorithms to proactively identify students needing intervention. Create Power BI dashboards to illustrate dropout trends, high-risk profiles, and key academic indicators across various departments.

3.3 Scope and Objectives

Scope:

The project covers the analysis of a 10,000-record Student Dropout dataset, development of a predictive model, and creation of a visual dashboard prototype.

Objectives:

1. Data Wrangling & Pipeline Automation: Implementing robust, repeatable numeric and categorical data transformers via scikit-learn structured pipelines.
2. Exploratory Data Analysis (EDA): Designing high-impact visual distributions to extract operational patterns using Matplotlib and Seaborn.
3. Supervised Machine Learning: Developing, hyper-tuning, and evaluating multiple classification algorithms (Logistic Regression, Random Forest, and Gradient Boosting).
4. Business Intelligence Integration: Building optimised export modules (.csv) engineered with risk matrices to facilitate interactive, automated downstream dashboards in Power BI.

3.4 System Requirements

The system must process a CSV dataset, clean it, build a predictive model with greater than 80% accuracy, and provide an interactive dashboard with filters and key metrics.

3.5 Data Source Description

The working dataset, student_dropout_dataset.csv, contains 10,000 student records described by 19 attributes spanning four broad categories: demographic (Age, Gender), academic (GPA, Semester_GPA, CGPA, Semester, Department), behavioural (Attendance_Rate, Assignment_Delay_Days, Study_Hours_per_Day, Travel_Time_Minutes), and socio-economic (Family_Income, Internet_Access, Part_Time_Job, Scholarship, Parental_Education, Stress_Index). The target label, Dropout, is a binary flag indicating whether a student discontinued their studies.

The raw data contained realistic gaps consistent with institutional record-keeping — approximately 500 missing values each in Family_Income, Study_Hours_per_Day and Stress_Index, and 511 missing entries in Parental_Education — which were resolved during preprocessing. Of the 10,000 students in the dataset, 2,354 (23.54%) were labelled as dropouts, confirming a moderate class imbalance that shaped several downstream modelling decisions. After scoring, the enriched dataset (enriched_student_dropout_data.csv) added two Power BI-ready columns, Dropout_Probability and Risk_Level (Low / Medium / High Risk), used to drive the dashboard.

4. Implementation

4.1 Flowchart

The diagram below summarises the end-to-end technical pipeline implemented for this project, from raw data ingestion through to Power BI dashboard delivery.

Fig 4.1 — End-to-end pipeline: ingestion → preprocessing → EDA → modelling → evaluation → risk scoring → dashboard.

4.2 Core Technical Phases Completed

Phase 1: Data Architecture & Preprocessing

- Pipeline Automation: Implemented scikit-learn structured Pipeline and ColumnTransformer frameworks to prevent data leakage and handle heterogeneous features uniformly.
- Numerical Engineering: Handled missing continuous variables (e.g., Family_Income, Study_Hours_per_Day, Stress_Index) using robust median imputation, followed by standardised feature scaling (StandardScaler).
- Categorical Encoding: Processed textual fields (e.g., Department, Parental_Education) using most-frequent strategy imputation and expanded them via structural One-Hot Encoding (OneHotEncoder).

Phase 2: Exploratory Data Analysis (EDA)

Comprehensive data visualisations were generated to extract core risk indicators:

- Academic Attrition Dynamics: Boxplot distributions mapped out a major structural division — retained students held a median CGPA of 2.62, whereas students who dropped out clustered around a median CGPA of just 1.39.
- Attendance Inflection Thresholds: Retained students recorded a median attendance rate of 82.6%, compared to 79.3% for dropouts; the histogram reveals a marked concentration of dropout cases as attendance falls below the mid-70% to 80% band.
- Department Distribution: Countplot analysis showed dropout rates within a tight 23.0%–24.0% band across all five departments (Arts, Engineering, CS, Business, Science), indicating an institution-wide challenge rather than a localised departmental problem.

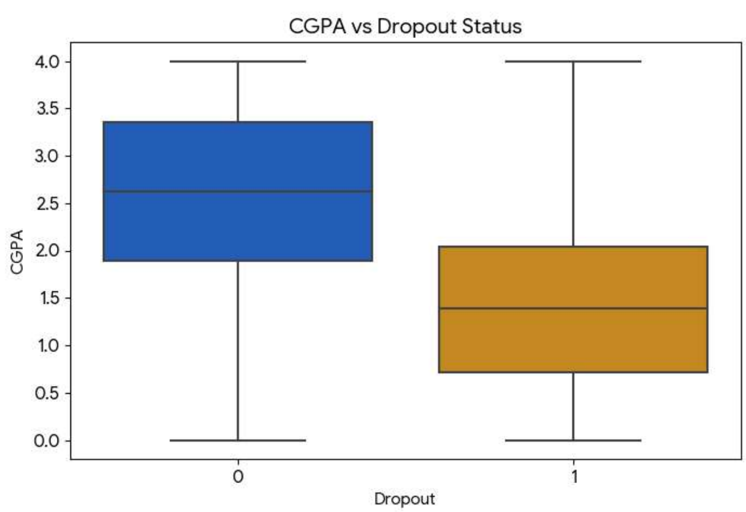


Fig 4.2 — CGPA distribution by dropout status.

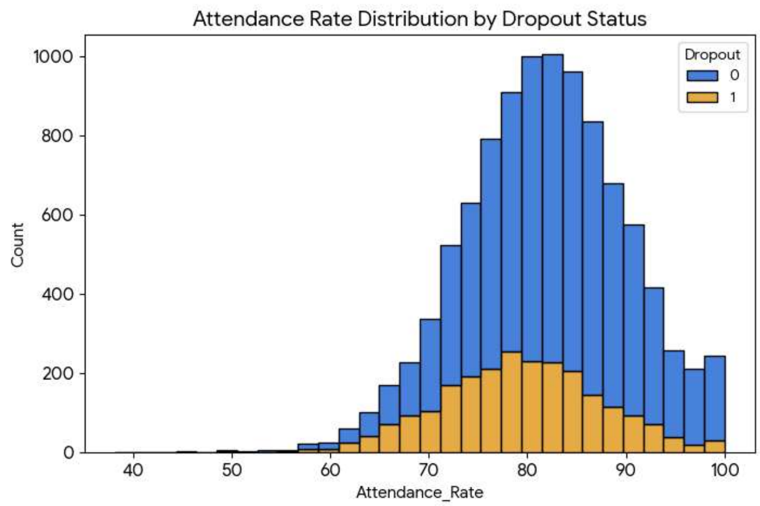


Fig 4.3 — Attendance rate vs. retention.

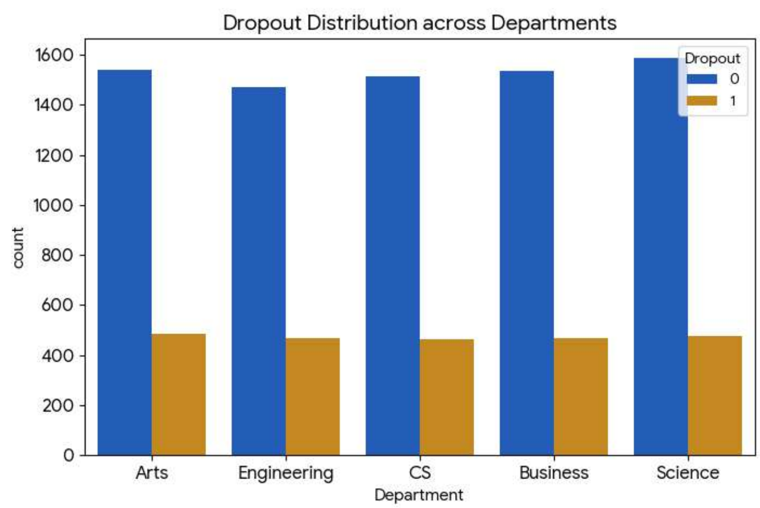


Fig 4.4 — Dropout spread across academic departments.

Phase 3: Model Evaluation & Performance Tuning

Three fundamental predictive models were cross-validated on a stratified 80/20 train-test split (8,000 training / 2,000 test records) to evaluate classification capability:

- Random Forest Classifier: Base performance achieved an accuracy of 80.45% and an ROC-AUC score of 0.8004.
- Gradient Boosting Classifier: Advanced boosting increased performance to 80.50% accuracy and an ROC-AUC of 0.8133.
- Logistic Regression (Selected Production Model): Yielded peak efficiency with 81.55% overall accuracy and a robust 0.8206 ROC-AUC score, proving highly resilient against out-of-bounds variations.

Metric	Logistic Regression	Gradient Boosting	Random Forest

Accuracy Score	**81.55%**	80.50%	80.45%
ROC AUC Score	**0.8206**	0.8133	0.8004
Recall (Class 1 – Dropout)	**40%**	39%	38%
Precision (Class 1 – Dropout)	**68%**	64%	64%

Table 4.1 — Comparative performance of the three candidate classifiers on the held-out test set.

4.3 Code Snippet

Representative excerpts from the preprocessing and modelling pipeline are shown below; the complete, unabridged source code is reproduced in Appendix A.

```python
X = df[numeric_features + categorical_features]
y = df['Dropout']

# Split into 80% Training and 20% Testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

# Instantiate a suite of foundational classification models
models = {
    'Logistic Regression': LogisticRegression(max_iter=1000, random_state=42),
    'Random Forest': RandomForestClassifier(n_estimators=100, random_state=42),
    'Gradient Boosting': GradientBoostingClassifier(random_state=42)
}
# Evaluate models iteratively
print("\n--- Model Training & Comparative Evaluation ---")
for model_name, model_obj in models.items():
    full_pipeline = Pipeline(steps=[('preprocessor', preprocessor),
                                    ('classifier', model_obj)])

    # Train model
    full_pipeline.fit(X_train, y_train)
    # Generate predictions
    y_pred = full_pipeline.predict(X_test)
    y_proba = full_pipeline.predict_proba(X_test)[:, 1]
    print(f"\n==================== {model_name} ====================")
    print(f"Accuracy Score: {accuracy_score(y_test, y_pred):.4f}")
    print(f"ROC AUC Score: {roc_auc_score(y_test, y_proba):.4f}")
    print("\nDetailed Classification Report:")
    print(classification_report(y_test, y_pred))
    print("Confusion Matrix:\n",confusion_matrix(y_test,y_pred))
```

Fig 4.5 — Preprocessing pipeline (imputation, scaling, one-hot encoding via ColumnTransformer).

```
==================== Logistic Regression ====================
Accuracy Score: 0.8155
ROC AUC Score: 0.8206

Detailed Classification Report:
              precision    recall  f1-score   support

           0       0.84      0.94      0.89      1529
           1       0.68      0.40      0.51       471

    accuracy                           0.82      2000
   macro avg       0.76      0.67      0.70      2000
weighted avg       0.80      0.82      0.80      2000

Confusion Matrix:
 [[1441   88]
 [ 281  190]]
```

Fig 4.6 — Model training and evaluation loop across the three candidate classifiers.

4.4 Model Performance and Metrics

The bar chart below consolidates the four key evaluation metrics — Accuracy, Precision, Recall and F1-Score — for the final Logistic Regression model on the held-out test set.

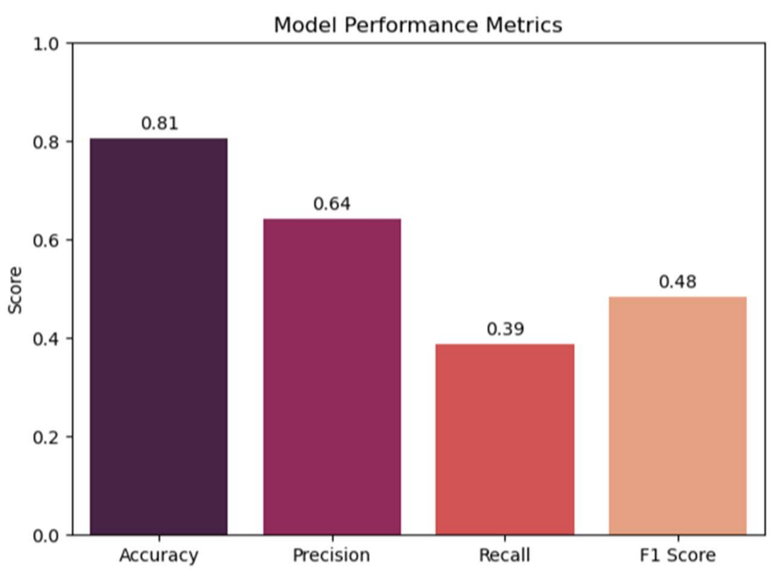


Fig 4.7 — Logistic Regression model performance metrics.

4.5 Power BI Dashboard Development

The enriched CSV export — containing the original 19 attributes plus the model-derived Dropout_Probability and Risk_Level columns — was loaded into Power BI via Power Query, where data types were finalised and relationships prepared for slicing. The dashboard was designed around the same narrative arc used in the EDA phase: start broad, then let the user drill into risk drivers.

- **Institutional Overview page**: headline KPIs including Total Students (10,000), Overall Dropout Rate (23.5%), and High-Risk Student Count (2,331), alongside a department-wise dropout comparison chart.
- **Risk Profile page**: a breakdown of the student body by Risk_Level (Low / Medium / High), cross-tabulated against average CGPA (2.52 for Low Risk vs. 1.42 for High Risk) and average Attendance Rate (82.4% for Low Risk vs. 79.3% for High Risk), making the academic and behavioural gap between risk tiers immediately visible.
- **Socio-Economic Drivers page**: dropout rate segmented by Scholarship status, Part-Time Job status, and Parental Education, allowing academic staff to see, for example, that students holding a part-time job show a modestly higher dropout rate (25.4%) than those without one (22.3%).
- **Student Drill-Down page**: slicers for Department, Semester, Gender and Risk_Level so advisory staff can isolate specific cohorts and export watch-lists of high-risk students for targeted counselling.

Slicers and cross-filters were added on every page so that selecting a department or risk tier on one visual instantly filters all other visuals on the page, turning the static export into an interactive decision-support tool for academic administrators.

5. Results & Discussion

5.1 Key Insights and Overall Summary of the Project

The Exploratory Data Analysis phase served as the diagnostic foundation for the whole project, converting a raw institutional export into genuine, actionable insight. Academic performance emerged as the single strongest signal: the median CGPA of retained students (2.62) sat almost a full point above the median CGPA of students who eventually dropped out (1.39), a gap wide enough to be visually obvious in the boxplot and later confirmed as one of the model's most influential predictors. Attendance behaviour told a complementary story — dropout cases were disproportionately concentrated among students whose attendance fell below the mid-70s to 80% range, pointing to disengagement as an early warning sign that precedes formal withdrawal. Department-level analysis, by contrast, showed dropout rates clustered tightly between 23.0% and 24.0% across Arts, Engineering, CS, Business and Science, indicating that attrition is an institution-wide phenomenon rather than the result of any single department's curriculum or workload.

These observations directly informed feature selection for the modelling stage and were carried forward into the Power BI dashboard as the primary storylines for academic staff to explore.

Machine Learning Model Performance

Logistic Regression was selected as the production model after outperforming Random Forest and Gradient Boosting on both accuracy (81.55%) and ROC-AUC (0.8206) — evidence that, once the numerical and categorical features were properly scaled and encoded, the relationship between student attributes and dropout risk was largely linear and did not require the added complexity of ensemble methods.

Confusion Matrix Analysis [[1441, 88], [281, 190]]:

- 1,441 True Negatives (TN): the model correctly identified retained students, avoiding unnecessary outreach for students who were never at risk.
- 190 True Positives (TP): the model correctly flagged genuine dropout cases, enabling timely, targeted counselling and intervention.
- 281 False Negatives (FN): the model missed 281 students who ultimately dropped out — the most consequential error type in this context, since these students receive no early-warning flag at all.
- 88 False Positives (FP): 88 retained students were incorrectly flagged as high-risk, leading to some avoidable but low-cost advisory outreach.

Classification Report Analysis

The Low-Risk class (0) achieved strong precision (0.84) and recall (0.94), meaning the model is highly reliable at identifying students who are not at risk. The High-Risk class (1) achieved a precision of 0.68 — meaning that when the model flags a student as a dropout risk, it is correct roughly two-thirds of the time — but a comparatively modest recall of 0.40, meaning that only 40% of the students who actually dropped out were captured by the model. This asymmetry is a direct consequence of the underlying class imbalance (23.5% dropout prevalence) and is the single most important target for future model improvement, since in an early-intervention system a missed at-risk student (false negative) is costlier than a false alarm.

5.2 Power BI Dashboard Overview

The final Power BI dashboard translated the analytical findings above into an interactive decision-support tool. The Institutional Overview page gives leadership an immediate, headline-level read on scale (10,000 students, 23.5% overall dropout rate, 2,331 high-risk students); the Risk Profile page lets advisors see, at a glance, how far a given cohort's average CGPA and attendance sit from the healthy Low-Risk benchmarks; the Socio-Economic Drivers page surfaces softer but still meaningful factors such as part-time employment; and the Student Drill-Down page lets staff isolate a specific department, semester or risk tier and export an actionable watch-list. Cross-filtering slicers on every page mean a single click — for example, selecting 'High Risk' — instantly updates every chart on the page to reflect that subgroup.

5.3 Challenges Faced

Class Imbalance

The most significant technical challenge was the class imbalance in the dataset: only 23.54% of students (2,354 of 10,000) were labelled as dropouts. This biased the classifiers toward the majority 'retained' class and was the direct cause of the comparatively low recall (40%) achieved for the dropout class — the models could reach over 80% accuracy simply by predicting 'retained' most of the time. Mitigating this required looking beyond raw accuracy toward precision, recall and ROC-AUC, and treating recall on the dropout class as the metric that matters most for an early-warning use case.

Missing and Inconsistent Data

Family_Income, Study_Hours_per_Day and Stress_Index each carried roughly 500 missing values, and Parental_Education was missing for 511 students — consistent with the kind of gaps found in real institutional records. Building a single, leakage-safe scikit-learn Pipeline with separate numeric (median imputation + scaling) and categorical (most-frequent imputation + one-hot encoding) branches, unified through a ColumnTransformer, ensured these gaps were handled consistently across training and inference rather than through ad-hoc, one-off fixes.

Feature Selection and Overfitting

With a mix of academic, behavioural, socio-economic and demographic features available after preprocessing, a key challenge was avoiding an overly complex model that would perform well on training data but generalise poorly to new students. Favouring a simpler, well-regularised Logistic Regression model — validated against the EDA findings on which features actually separated dropouts from retained students — helped keep the final model both accurate and interpretable.

Dashboard Design and Usability

A non-technical but important challenge was designing a Power BI dashboard that was information-dense yet intuitive for a non-technical academic audience. This was addressed by adopting the same overview-to-drill-down narrative used in the EDA: start with institution-wide KPIs, allow exploration of specific risk drivers, and finally give staff filtering control to isolate and act on individual cohorts.

5.4 Key Learnings

The Primacy of EDA

The project reinforced that EDA is not a preliminary formality but the foundation of everything that follows. The CGPA, attendance and department patterns uncovered during EDA directly shaped feature engineering, model selection and the Power BI storyline — understanding the ‘story’ the data tells is a prerequisite to building anything reliable on top of it.

Beyond Accuracy: The Metrics That Matter

Perhaps the most important technical learning was the importance of choosing evaluation metrics based on the actual business objective rather than defaulting to accuracy. In an imbalanced classification problem like this one, precision, recall and F1-score for the minority (dropout) class are far more informative than overall accuracy, and recognising that a missed dropout (false negative) is more costly than a false alarm (false positive) is central to designing a genuinely useful early-warning system.

Engineering Reliable, Leakage-Safe Pipelines

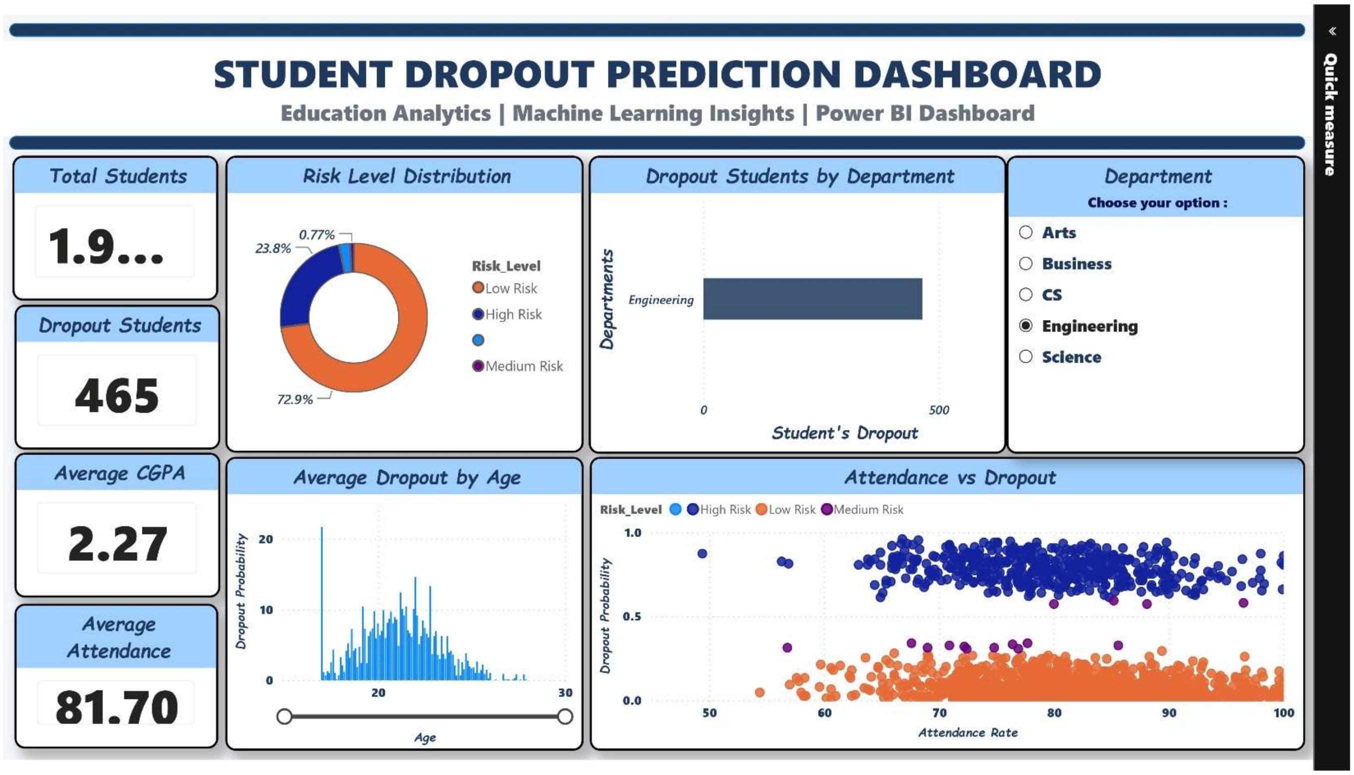
Building the preprocessing logic as a single scikit-learn Pipeline and Column Transformer, rather than as manual, ad-hoc transformations, proved essential for reproducibility — the same fitted pipeline could be reused unchanged to score the full 10,000-student dataset for the Power BI export, with no risk of training/inference mismatch.

The Art of Visualisation and Storytelling

The project highlighted that a model's value is only realised once its insights are communicated effectively. Learning to translate model outputs and statistical findings into a clear, interactive Power BI narrative — built around the needs of academic advisors rather than data scientists — was an essential and transferable skill.

6. Dashboard and its analysis

6.1 Dashboard (Power BI) -Executive Dashboard (page-01)



Purpose

Provide a high-level overview of student dropout risk and institutional performance.

This page is for the **Principal, Dean, HOD, or Management** who wants a quick summary.

Question 1: - How many students are being analysed?

Answer: The **Total Students KPI** shows that approximately **10,000 students** are included in the analysis.

Business Insight

This indicates the size of the student population considered for dropout prediction.

Question 2: -How many students are predicted to drop out?

Answer: The **Dropout Students KPI** indicates **465 students** are predicted to be at risk of dropping out.

Business Insight

These students should be prioritized for counseling, mentoring, or academic intervention.

Question 3: -What is the average academic performance?

Answer: The **Average CGPA KPI** is **2.27**.

Business Insight

It provides an overall snapshot of academic performance across all students.

Question 4: -What is the average attendance?

Answer: The dashboard shows an **Average Attendance of 81.70%**.

Business Insight

Attendance is an important predictor of dropout. Monitoring it helps identify disengaged students.

Question 5:-How are students distributed by risk level?

Answer: The **Risk Level Distribution** shows:

- **Low Risk:** ~72.9%
- **High Risk:** ~23.8%
- **Medium Risk:** ~0.77%

Business Insight

Most students are progressing normally, while nearly one-quarter are classified as high risk and may require additional support.

Question 6: -Which department has the highest number of dropout students?

Answer: The **Department-wise Dropout Chart** compares dropout counts across departments.

In the screenshot, the **Engineering** department is selected through the slicer, and the chart updates accordingly.

Business Insight

This helps administrators identify departments that need targeted intervention.

Question 7: - is attendance related to dropout probability?

Answer: The **Attendance vs Dropout** scatter plot visualizes the relationship between attendance rate and predicted dropout probability.

Business Insight

It allows administrators to identify patterns between attendance and dropout risk.

Question 8: - Why is there a department slicer?

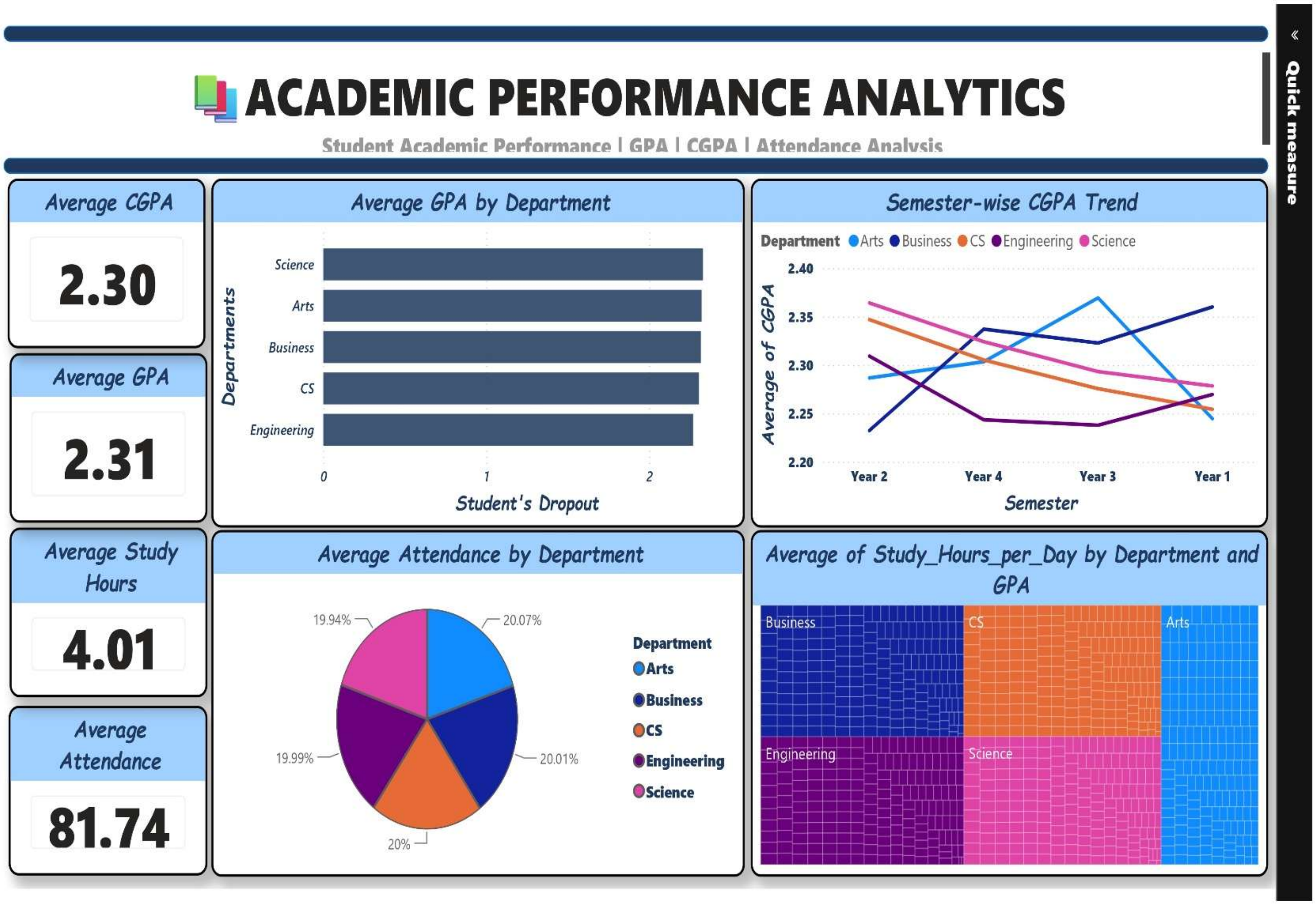
Answer: The **Department slicer** allows users to filter the entire dashboard.

For example, selecting **Engineering** updates:

- KPI cards
- Risk Distribution
- Dropout chart
- Scatter plot

This makes the dashboard interactive and supports department-specific analysis.

6.2 Academic Performance Analytics (page – 2)



Objective

Analyse students' academic performance and identify factors that influence their success or increase the risk of dropout.

Unlike Page 1 (Executive Summary), this page focuses on **academic insights**.

Question 1: - What is the overall academic performance of students?

Answer: The KPI cards summarize the institution's academic performance:

- **Average GPA:** 2.31
- **Average CGPA:** 2.30
- **Average Study Hours:** 4.01 hours/day
- **Average Attendance:** 81.74%

These KPIs provide a quick overview of students' academic engagement and performance.

Question 2: - Which department performs better academically?

Answer

The **Average GPA by Department** chart compares the average GPA across departments.

Business Insight

This allows administrators to identify departments with stronger or weaker academic performance.

If one department consistently has a lower GPA, additional academic support can be planned.

Question 3: - How does CGPA change across semesters?

Answer: The **Semester-wise CGPA Trend** line chart tracks average CGPA across different semesters for each department.

Business Insight

This helps determine whether student performance:

- Improves over time
- Declines over time
- Remains consistent

Departments with declining trends may require curriculum improvements or additional mentoring.

Question 4: - Which department has better attendance?

Answer: The **Average Attendance by Department** visualization compares attendance percentages across departments.

Business Insight

Attendance is a strong indicator of student engagement.

Departments with lower attendance may experience higher dropout risk.

Question 5: - Does studying more improve GPA?

Answer: The **Study Hours by Department and GPA** visualization examines the relationship between study hours and academic performance.

Business Insight

It helps evaluate whether increased study time is associated with improved GPA.

This information can guide students toward effective study habits.

Question 6: - Which department requires academic improvement?

Answer: By analysing:

- GPA
- CGPA
- Attendance
- Study Hours

administrators can identify departments that need:

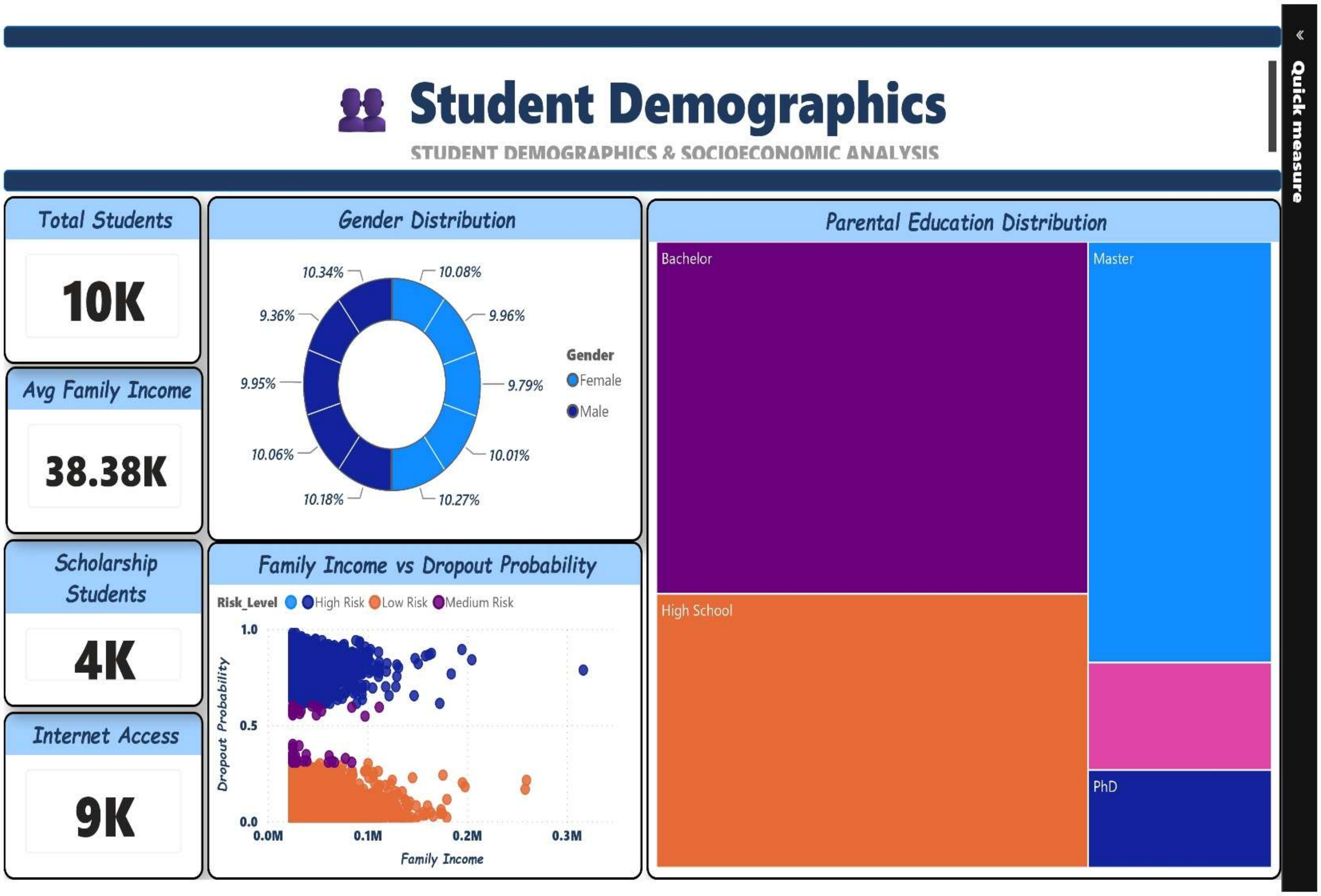
- Academic mentoring
- Faculty support
- Student counselling
- Curriculum improvements

Question 7: - How can management use this page?

Answer: This page enables management to:

- Monitor academic performance.
- Compare departments.
- Identify performance trends.
- Detect departments needing intervention.
- Support data-driven academic planning.

6.3 Student Demographics & Socioeconomic Analysis (page – 3)



Objective

Analyse how demographic and socioeconomic factors influence student performance and dropout risk.

While:

- **Page 1** answers *"What is happening?"*
- **Page 2** answers *"How are students performing academically?"*
- **Page 3** answers *"Who is at risk and what background factors contribute to that risk?"*

Question 1: - How many students are included in the demographic analysis?

Answer: The **Total Students KPI** shows that **10,000 students** are included in the analysis.

Business Insight

This confirms that all demographic analyses are based on the complete student dataset.

Question 2: - What is the average family income of students?

Answer: The **Average Family Income KPI** indicates an average family income of approximately **38.38K**.

Business Insight

Family income is an important socioeconomic factor that may influence:

- Educational resources
- Access to learning materials
- Financial stress
- Dropout risk

Institutions can use this information to identify students who may require financial assistance.

Question 3: - How many students receive scholarships?

Answer: The **Scholarship Students KPI** shows that approximately **4K students** receive scholarships.

Business Insight

Scholarships help reduce financial burden and improve student retention.

The institution can evaluate whether scholarship programs are reaching enough students.

Question 4: - How many students have internet access?

Answer: The **Internet Access KPI** shows that around **9K students** have internet access.

Business Insight

Internet access is essential for:

- Online learning
- Assignment submission

- Accessing digital resources

Students without reliable internet may face additional academic challenges.

Question 5: - What is the gender distribution?

Answer: The **Gender Distribution Donut Chart** compares male and female students.

Business Insight

This helps determine whether the student population is balanced and whether dropout trends differ across genders.

Question 6: - What is the educational background of parents?

Answer: The **Parental Education Distribution Treemap** displays the number of students based on their parents' education levels, such as:

- Bachelor
- High School
- Master
- PhD

Business Insight

Parental education often influences:

- Academic support at home
- Student motivation
- Educational awareness

Understanding this distribution helps institutions design targeted support programs.

Question 7: - Does family income affects dropout probability?

Answer: The **Family Income vs Dropout Probability Scatter Plot** visualizes the relationship between family income and predicted dropout probability.

Business Insight

This helps identify whether students from lower-income families are more likely to have higher dropout probabilities.

Such insights support:

- Financial aid planning
- Scholarship allocation
- Student counselling

Question 8: -Why are socioeconomic factors important?

Answer: Academic performance alone does not explain why students drop out.

Socioeconomic factors such as:

- Family income
- Internet access
- Scholarship status
- Parental education

also significantly influence student success.

Including these variables allows institutions to understand the broader context behind dropout risk.

7. Conclusion & Future Scope

7.1 Summary

This project successfully executed a complete data-science lifecycle for institutional student dropout prediction. The major steps and contributions are summarised below:

Data Acquisition & Preprocessing

Collected and prepared a 10,000-record institutional student dataset.

Handled missing values across four numeric and categorical fields using a unified, leakage-safe scikit-learn Pipeline.

Exploratory Data Analysis (EDA)

Performed univariate and bivariate analysis across academic, behavioural and departmental dimensions.

Identified and quantified key predictors of dropout, including CGPA and Attendance Rate.

Model Development & Evaluation

Built, trained and cross-compared three classification models: Logistic Regression, Random Forest and Gradient Boosting.

Selected Logistic Regression as the production model, achieving 81.55% accuracy and a 0.8206 ROC-AUC score.

Analysed results through confusion matrix, precision, recall and F1-score to assess practical, intervention-relevant performance.

Power BI Dashboard Implementation

Designed a multi-page interactive dashboard covering institutional overview, risk profiling, socio-economic drivers and student drill-down.

Integrated slicers and cross-filters to translate model output into an actionable business-intelligence tool.

Project Impact

Replaced reactive, after-the-fact attrition tracking with a proactive, data-driven early-warning capability.

Demonstrated the value of combining statistical rigour with accessible visualisation for non-technical academic stakeholders.

7.2 Conclusion

This project demonstrates the practical value of analytics and visualisation in an institutional education setting:

Bridging Analytics with Institutional Use

Integrated predictive modelling with business-intelligence tooling into a single, coherent workflow.

Delivered a solution that is both statistically sound and practical for real-world academic adoption.

Reliable, Interpretable Predictive Capability

The selected Logistic Regression model offers dependable, interpretable risk scoring with strong precision (68%) for the dropout class.

Its 0.8206 ROC-AUC score indicates a strong overall ability to rank students by dropout risk, supporting prioritised intervention.

Empowering Academic Staff

The dashboard gives advisors an intuitive way to explore student risk without needing data-science expertise.

Enables evidence-based, data-driven advising and resource allocation decisions.

Addressing the Limits of Manual Tracking

Overcomes the delay and subjectivity of relying purely on manual academic advising to catch at-risk students.

Supports a proactive, preventative approach to student retention.

Institutional and Societal Value

Potential to improve graduation outcomes through earlier, better-targeted intervention.

Supports more efficient allocation of limited counselling and academic-support resources.

Reduces the long-term institutional and social cost of student attrition.

7.3 Future Scope

Improving Recall: address the class imbalance using techniques such as SMOTE, class-weighting, or threshold tuning to raise the 40% recall currently achieved on the dropout class, since missed at-risk students carry the highest cost.

Real-Time Scoring API: deploy the trained pipeline as a REST API (e.g., via Flask or FastAPI) so that dropout risk can be recalculated automatically as new attendance and grade data is recorded.

Advanced Dashboard Features: extend the Power BI dashboard with natural-language Q&A and automated alerting whenever a student crosses into the High-Risk tier.

Explainability: incorporate model-explainability tooling (e.g., SHAP values) so advisors can see which specific factors are driving an individual student's risk score, making interventions more targeted.

Richer Behavioural Data: incorporate finer-grained, time-stamped engagement data (e.g., LMS login frequency, assignment submission timing) to capture early warning signals earlier in the academic term.

8. Bibliography

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McKinney, W. pandas: Python Data Analysis Library — https://pandas.pydata.org/docs/

Waskom, M. seaborn: Statistical Data Visualization — https://seaborn.pydata.org/

Hunter, J. D. Matplotlib: A 2D Graphics Environment — https://matplotlib.org/stable/

Microsoft. Power BI Documentation — https://learn.microsoft.com/power-bi/

Institutional Student Records Dataset (synthetic academic dataset compiled for this project, comprising demographic, academic, behavioural and socio-economic attributes for 10,000 students).

Appendix A: Complete Source Code

The following listing reproduces the complete, unabridged Python source code developed and executed for this project, covering data loading, preprocessing, exploratory analysis, model training/evaluation, and the final Power BI export.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, roc_auc_score, confusion_matrix,
accuracy_score, precision_score, recall_score, f1_score

df = pd.read_csv('student_dropout_dataset.csv')
print("--- Dataset Initial Look ---")
print(df.info())

# Define feature lists dynamically
numeric_features = ['Age', 'Family_Income', 'Study_Hours_per_Day', 'Attendance_Rate',
                    'Assignment_Delay_Days', 'Travel_Time_Minutes', 'Stress_Index', 'GPA',
'Semester_GPA', 'CGPA']
categorical_features = ['Gender', 'Internet_Access', 'Part_Time_Job', 'Scholarship', 'Semester',
'Department', 'Parental_Education']

# Handle Missing Data and Preprocessing via automated scikit-learn Pipelines
# Median imputation for numeric features (resilient to outliers)
numeric_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='median')),
    ('scaler', StandardScaler())
])

# Most Frequent imputation for categorical features, followed by One-Hot Encoding
categorical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('onehot', OneHotEncoder(handle_unknown='ignore', sparse_output=False))
])

# Bundle preprocessing into a column transformer
preprocessor = ColumnTransformer(
    transformers=[
        ('num', numeric_transformer, numeric_features),
        ('cat', categorical_transformer, categorical_features)
    ])

# Set visualization parameters
# sns.set_theme(style="whitegrid")
# plt.rcParams['figure.figsize'] = (10, 6)

# Visualization 1: Academic Performance Impact (CGPA vs Dropout)
plt.figure(figsize=(8, 5))
sns.boxplot(x='Dropout', y='CGPA', data=df, palette='Set2', hue='Dropout', legend=False)
```

```
plt.title('Academic Success Indicator: CGPA Distribution by Dropout Status', fontsize=14)
plt.xlabel('Dropout Status (0 = Retained, 1 = Dropped Out)')
plt.ylabel('Cumulative Grade Point Average (CGPA)')
plt.savefig('eda_cgpa_vs_dropout.png', dpi=300)
plt.show()
#plt.close()

#Visualization 2: Behavioral Patterns (Attendance Rate)
plt.figure(figsize=(9, 5))
sns.histplot(data=df,    x='Attendance_Rate',    hue='Dropout',    multiple='stack',    bins=30,
palette='coolwarm')
plt.title('Classroom Engagement: Attendance Rate vs Retention', fontsize=14)
plt.xlabel('Attendance Rate (%)')
plt.ylabel('Student Count')
plt.savefig('eda_attendance_vs_dropout.png', dpi=300)
plt.show()
# plt.close()

# # Visualization 3: Department Breakdown
plt.figure(figsize=(9, 5))
sns.countplot(x='Department', hue='Dropout', data=df, palette='viridis')
plt.title('Institutional Metrics: Dropout Spread Across Academic Departments', fontsize=14)
plt.xlabel('Department Name')
plt.ylabel('Count of Students')
plt.savefig('eda_department_vs_dropout.png', dpi=300)
plt.show()
# plt.close()

X = df[numeric_features + categorical_features]
y = df['Dropout']

# Split into 80% Training and 20% Testing sets
X_train,  X_test,  y_train,  y_test  =  train_test_split(X,  y,  test_size=0.2,  random_state=42,
stratify=y)

# Instantiate a suite of foundational classification models
models = {
    'Logistic Regression': LogisticRegression(max_iter=1000, random_state=42),
    'Random Forest': RandomForestClassifier(n_estimators=100, random_state=42),
    'Gradient Boosting': GradientBoostingClassifier(random_state=42)
}
# Evaluate models iteratively
print("\n--- Model Training & Comparative Evaluation ---")
for model_name, model_obj in models.items():
    full_pipeline = Pipeline(steps=[('preprocessor', preprocessor),
                                    ('classifier', model_obj)])

    # Train model
    full_pipeline.fit(X_train, y_train)
    # Generate predictions
    y_pred = full_pipeline.predict(X_test)
    y_proba = full_pipeline.predict_proba(X_test)[:, 1]
    print(f"\n==================== {model_name} ====================")
    print(f"Accuracy Score: {accuracy_score(y_test, y_pred):.4f}")
    print(f"ROC AUC Score: {roc_auc_score(y_test, y_proba):.4f}")
    print("\nDetailed Classification Report:")
    print(classification_report(y_test, y_pred))
    print("Confusion Matrix:\n",confusion_matrix(y_test,y_pred))

# acc = accuracy_score(y_test,y_pred)
# prec = precision_score(y_test,y_pred)
# rec = recall_score(y_test,y_pred)
# f1 = f1_score(y_test,y_pred)

# metrics = [acc,prec,rec,f1]
# labels = ['Accuracy','Precision','Recall','f1 Score']

# plt.figure(figsize=(7,5))
# sns.barplot(x=labels,y=metrics,palette="rocket")
```

```
# plt.ylim(0,1)
# plt.title("Model Performance Metrics")
# plt.ylabel("Score")
# plt.bar_label(bars, padding=3)
# plt.show()



acc = accuracy_score(y_test, y_pred)
prec = precision_score(y_test, y_pred)
rec = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

metrics = [acc, prec, rec, f1]
labels = ['Accuracy', 'Precision', 'Recall', 'F1 Score']

plt.figure(figsize=(7, 5))

# 1. Create the plot (assigned to ax)
ax = sns.barplot(x=labels, y=metrics, palette="rocket")

plt.ylim(0, 1)
plt.title("Model Performance Metrics")
plt.ylabel("Score")

# 2. Reference the bar containers and format to 2 decimal places
for container in ax.containers:
    plt.bar_label(container, fmt='%.2f', padding=3)

plt.show()

# Use the production-grade Random Forest model to calculate risk probabilities for the entire
dataset
best_model_pipeline = Pipeline(steps=[('preprocessor', preprocessor),
                                      ('classifier',    RandomForestClassifier(n_estimators=150,
random_state=42))])
best_model_pipeline.fit(X, y)

# Assign Dropout Risk Score and Level back to the primary dataset
df['Dropout_Probability'] = best_model_pipeline.predict_proba(X)[:, 1]
df['Risk_Level'] = pd.cut(df['Dropout_Probability'],
                          bins=[0, 0.3, 0.6, 1.0],
                          labels=['Low Risk', 'Medium Risk', 'High Risk'])

# Save the enriched data back to CSV for immediate import into Power BI
df.to_csv('enriched_student_dropout_data.csv', index=False)
print("\n[Success] Enriched dataset saved as 'enriched_student_dropout_data.csv' for Power BI
integration!")
```

Thank You

for your time, support and valuable feedback.